NEOS TIMELINE

# From starter template to structured OS shell

This PDF is the public-facing process log for the NEOS rebuild. It is written to be readable by both technical and non-technical reviewers. Each milestone explains what changed, why it changed, and what the UI looked like at that step.

Current coverage: Stage 1A (original) → SHELL-01A/01B (v2 desktop shell) → SITE-SHELL-01 (liquid glass root) → SITE-SHELL-01B (WebGL2 refraction renderer) → SITE-SHELL-02 (hard reset).

## Project Snapshot

On 2026-03-16 the surface layer was hard reset to a blank canvas. Multiple AI agents iterating on the liquid glass surface in parallel created churn without convergence. The decision was made to rebuild incrementally with coordinated planning via agentchattr.

- Active lane: SITE-SHELL — rebuilding from blank canvas.
- Glass module preserved in `src/site/glass/` for re-integration.
- Phase 2 (deferred): ME-DESKTOP — OS desktop wired into ME.EXE.
- CONNECT-IMPLEMENTATION is frozen.
- The SHELL-01A/01B OS desktop code is preserved for ME-DESKTOP.

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### Timeline Map

1. Stage 1A (original): First shell UI pass — liquid glass channel navigation
2. SHELL-01A/01B (v2 build): OS desktop shell frame — re-scoped to ME.EXE
3. SITE-SHELL-01: Liquid glass channel surface mounted as site root
4. SITE-SHELL-01B: WebGL2 liquid glass renderer — reference-calibrated refraction, warm neutral palette
5. SITE-SHELL-02: Hard reset — surface stripped to blank canvas, coordinated rebuild via agentchattr (current)
6. ME-DESKTOP: OS desktop wired into ME.EXE (Phase 2)
7. Other channels: YOU.EXE, THIRD.EXE (Phase 3)

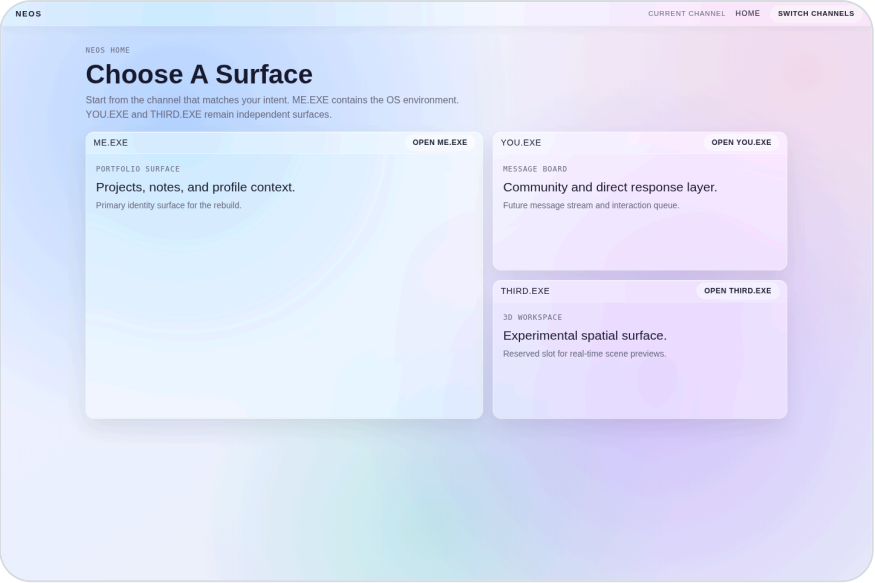
STAGE 1A — ORIGINAL

# First Shell UI Pass

This milestone strengthened the HOME landing composition while preserving the scoped ME.EXE OS boundary.

HOME presented a hero-led channel surface layout, and channel switching moved to a secondary quick-switch overlay. ME.EXE owned taskbar and launcher locally.

In plain English: landing felt deliberate and substantial without collapsing into a tile-browser or smart-TV style menu.




Stage 1A: HOME channel navigation surface with hero composition and secondary channel switcher. ME.EXE owned the taskbar in its own channel.

IMPORTANT CODE — STAGE 1A

## HOME Composition + Secondary Switcher

```
export function SiteShell(): ReactElement {
  return (
    <main className="neosShell">
      <div className="wallpaper" aria-hidden="true" />
      <ChannelBar ... onOpenSwitcher={() => setSwitcherOpen(true)} />
      <div className="channelViewport">
        <ChannelView channel={activeChannel} onNavigate=
{setActiveChannel} />
      </div>
      {switcherOpen ? <ChannelSwitcher ... /> : null}
      {bootVisible ? <BootOverlay /> : null}
    </main>
  );
}
```

This kept channel switching available but secondary, while preserving ME.EXE as the contained OS environment.

SHELL-01A — V2 REWRITE

# Desktop Shell Frame + Boot Gate + Static Taskbar

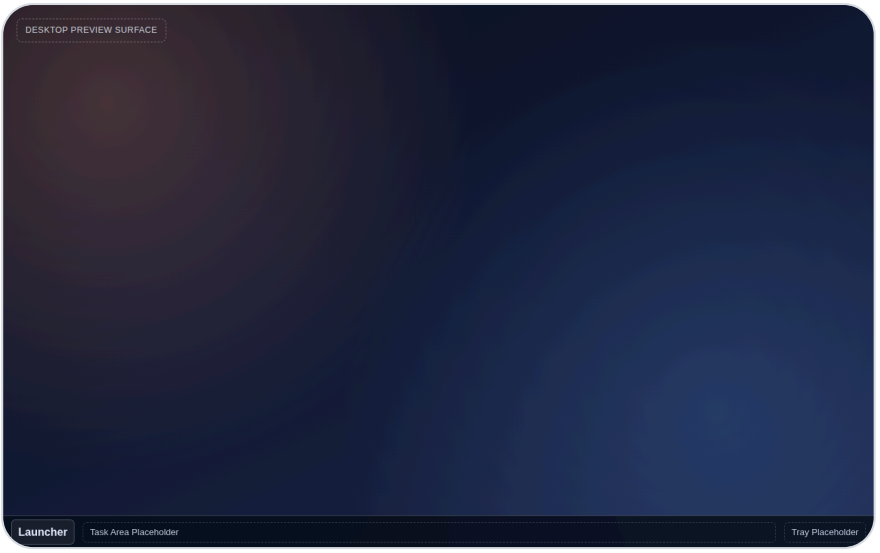
After the v2 workflow reset, the project moved to a contract-first desktop OS rewrite. SHELL-01A is the first milestone of the new architecture: a proper desktop shell frame with correct layer ordering, a boot gate, and a static taskbar.

This is a deliberate reset — not a visual upgrade of Stage 1A, but a structural rewrite with a new architecture. The channel navigation from Stage 1A is preserved in the timeline as history.

## What this established

- A stable desktop root in `Desktop.tsx` with spec-correct layer order.
- A contract-shaped desktop store implementing the full `DesktopStore` interface.
- A dedicated `WindowLayer` mount surface with no runtime logic.
- A non-blocking boot gate that does not wait on VFS readiness.
- `Z_LAYER` and `SHELL_Z_INDEX` constants locked in code.

Boot sequence: a fullscreen terminal-style overlay that dismisses on click or keypress. Fires immediately on load without blocking VFS or app services.




Desktop shell after boot is dismissed: gradient wallpaper, a "Desktop Preview Surface" label marking the future icon layer, and a static taskbar with Launcher, Task Area, and Tray placeholder zones.

IMPORTANT CODE — SHELL-01A

## Desktop Layer Stack

```
export function Desktop(): JSX.Element {
  const bootComplete = useDesktopStore((store) => store.bootComplete);

  return (
    <main aria-label="Desktop shell" style={desktopStyle}>
      <Wallpaper />
      <div aria-hidden="true" style={previewSurfaceStyle}>
        <span style={previewBadgeStyle}>Desktop Preview Surface</span>
      </div>
      <WindowLayer />
      <Taskbar />
      {!bootComplete ? <BootSequence /> : null}
    </main>
  );
}
```

This stack defines the layer order in code: wallpaper at the base, preview surface, window mount point, taskbar above, and boot overlay on top. Every z-index is driven by named constants, not magic numbers.

SITE-SHELL-01

## Liquid Glass Channel Surface Root

This milestone rewired the app root from OS desktop boot to a light, transparent channel surface.

`src/main.tsx` now mounts `SiteShell` from `src/site/`. The site root shows ME.EXE, YOU.EXE, and THIRD.EXE entries as placeholder channel panels, while runtime internals remain isolated and deferred.

SITE-SHELL-01 entry overlay on the light liquid glass site shell

Site entry overlay: lightweight branded entry moment, distinct from OS boot sequence.

SITE-SHELL-01 channel surface with ME.EXE, YOU.EXE, and THIRD.EXE entries

Liquid glass channel surface at root level, with isolated channel entry cards and no site-level desktop runtime.

IMPORTANT CODE — SITE-SHELL-01

### Root Mount Rewire

```
import { SiteShell } from './site/SiteShell';

ReactDOM.createRoot(document.getElementById('root')!).render(
  <React.StrictMode>
    <SiteShell />
  </React.StrictMode>
);
```

This isolates desktop/windowing code from the site root and restores channel-surface-first navigation as the primary entry model.

## What Was Intentionally Deferred in SHELL-01A

- No launcher behavior in this milestone.
- No notification or context-menu systems yet.
- No desktop icon interactions yet.
- No live app windows inside `WindowLayer` yet.
- No CONNECT gameplay or runtime implementation.

SITE-SHELL-01B

## WebGL2 Liquid Glass Renderer

A raw WebGL2 refraction renderer was added beneath the site shell panels. The glass effect is physically grounded — Snell's law refraction, Poisson-disk blur, specular rim lighting, and a convex squircle surface profile matching the archisvaze/liquid-glass reference.

Parameters were calibrated to the reference defaults: IOR 3.0, blur 2 px (nearly clear), bezel 60 px, specular 0.5. The surface profile was updated from a simple power curve to the reference's convex squircle function.

The blue colour palette was replaced with warm neutrals. Buttons became nearly-transparent glass panels with backdrop blur rather than opaque white fills.

### What this established

- A single full-screen WebGL2 canvas rendering all glass regions in one shader pass.
- A glass registry and React hook for components to self-register their bounding rects.
- Zero new runtime dependencies — raw WebGL2 only.
- Visual parity with the archisvaze/liquid-glass reference implementation.
- Warm neutral colour palette (no blue) throughout the site shell.

IMPORTANT CODE — SITE-SHELL-01B

## Convex Squircle Surface Profile (GLSL)

```
// Matches archisvaze/liquid-glass reference
float surfaceHeight(float t) {
  return pow(1.0 - pow(1.0 - t, 4.0), 0.25);
}
```

This surface profile produces the characteristic lens-like bulge at the glass edges with a relatively flat interior — matching the reference's natural liquid glass appearance.

SITE-SHELL-02

# Hard Reset: Blank Canvas

After SITE-SHELL-01B, multiple AI agents (Claude, Codex, Gemini) iterated on the home page surface in parallel. The work included implementing a 3-tier material hierarchy, hero card layouts, live wallpaper switching, and repeated glass parameter tuning. Each iteration was well-built individually, but the agents kept pushing competing visual directions without converging.

The SiteShell and ChannelSurface files grew to 135 and 517 lines of dense inline styles. Coordinating changes across agents became harder than the original implementation. The decision was made to hard reset the surface layer and rebuild incrementally with agent coordination through agentchattr.

## What was stripped

- `SiteShell.tsx` — gutted to an empty `<main>` with warm background.
- `ChannelSurface.tsx` — gutted to an empty `<section>`.
- `styles.css` — stripped to base CSS resets (~800 lines of component styles removed).
- `App.tsx` — deleted (unused duplicate entry point).

## What was preserved

- Entire `src/site/glass/` WebGL2 module (6 files) for future re-integration.
- `siteStore.ts`, `SiteBootOverlay.tsx` — infrastructure components.
- All windowing, app, core, and service code untouched.
- All docs, specs, config, and build tooling intact.

In plain English: we wiped the screen clean because too many cooks were adjusting the recipe at the same time. The kitchen equipment is all still there — we just need to cook one dish at a time with everyone agreeing on the menu first.

## Next Milestone Focus

Rebuilding will proceed one piece at a time with coordinated planning via agentchattr. The glass module will be re-integrated once the layout foundation is stable.

- SITE-SHELL-03 — Navigation bar rebuild.
- SITE-SHELL-04 — Channel routing and content surfaces.
- SITE-SHELL-05 — Glass re-integration.
- ME-DESKTOP-01 — OS desktop wired into ME.EXE (Phase 2).